RNAseq

Michael Schatz

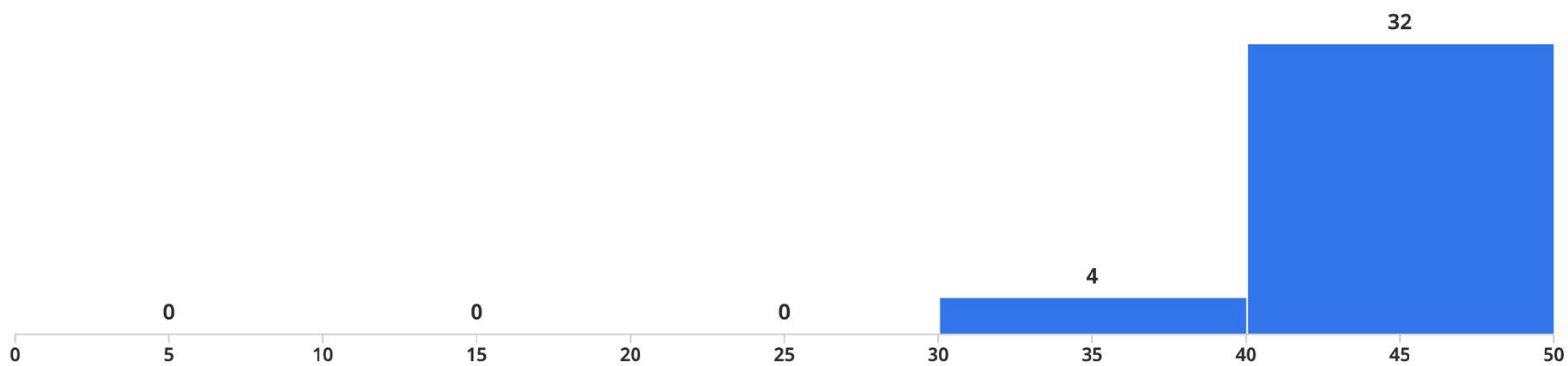
October 7, 2024

Lecture 12. Applied Comparative Genomics

Exam 1 Review

Review Grades for Exam 1

● Regrade Requests Open ● Grades Published



Minimum	Median	Maximum	Mean	Std Dev
30.0	44.5	49.0	43.54	3.96

Project Proposal
Due Wednesday Oct 15 @ 11:59pm

Project Proposal

Assignment Date: Monday October 6, 2025
Due Date: Wednesday, October 15 2025 @ 11:59pm

Review the [Project Ideas] page

Work solo or form a team for your class project of no more than 3 people.

The proposal should have the following components:

- Name of your team
- List of team members and email addresses
- Short title for your proposal
- 1 to 2 paragraph description of what you hope to do and how you will do it
- References to 3 to 4 relevant papers
- References/URLs to datasets that you will be studying (Note you can also use simulated data)
- Please add a note if you need me to sponsor you for a MARCC account (high RAM, GPUs, many cores, etc)
- Please add a note if you need me to sponsor you for an AnVIL cloud billing account (dynamic resources)

Submit the proposal as a 1 to 2 page PDF on GradeScope (each team should submit one proposal and tag all people in the team). After submitting your proposal, I will provide feedback. If necessary, we can schedule a time to discuss your proposal, especially to ensure you have access to the data that you need. The sooner that you submit your proposal, the sooner we can schedule the meeting.

Later, you will present your project in class during the last week of class. You will also submit a written report (5-7 pages) of your project, formatting as a Bioinformatics article (Intro, Methods, Results, Discussion, References). Word and LaTeX templates are available at https://academic.oup.com/bioinformatics/pages/submission_online

Please use Piazza to coordinate proposal plans!

"AlexNet"

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

Abstract

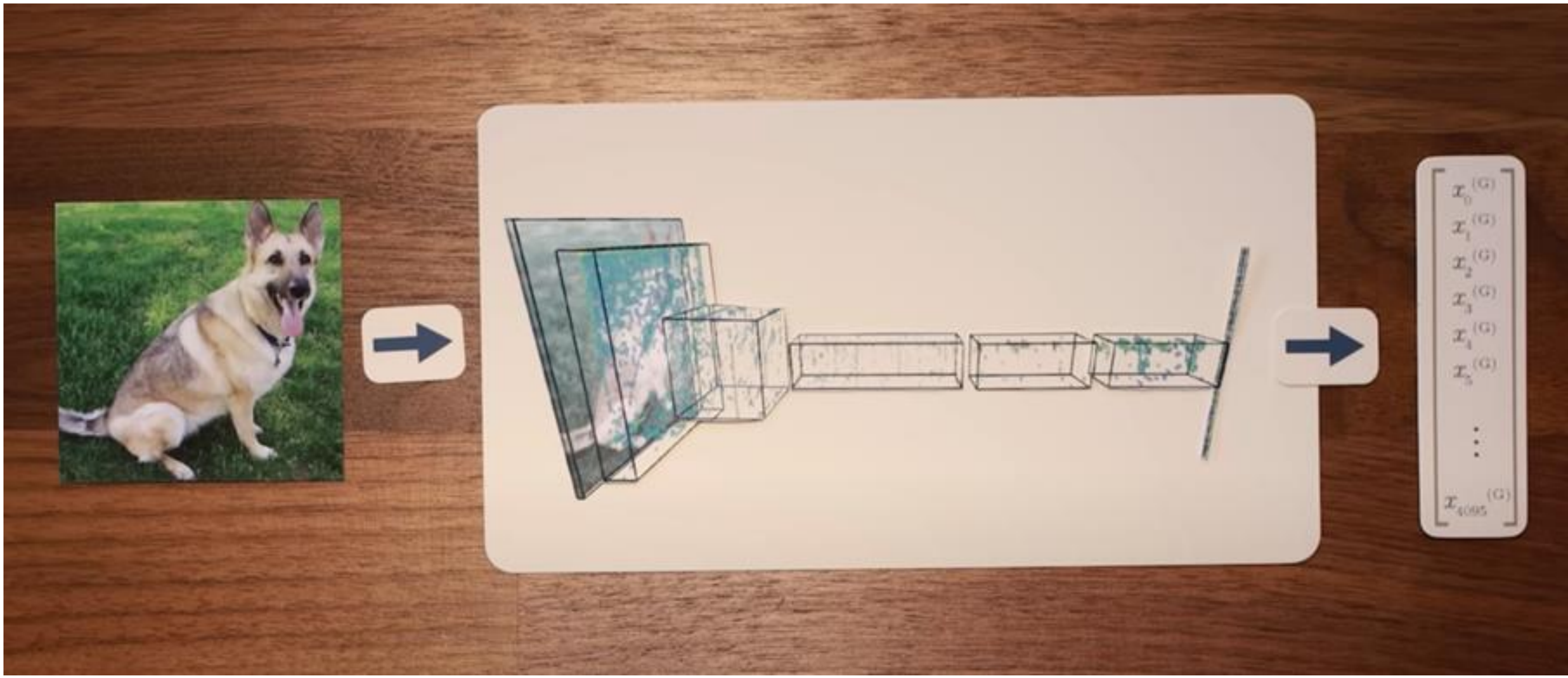
We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

Alex Krizhevsky
U. Toronto/Google

Ilya Sutskever
U. Toronto/OpenAI/
Safe Superintelligence Inc

Geoffrey Hinton
U. Toronto/Vector Institute

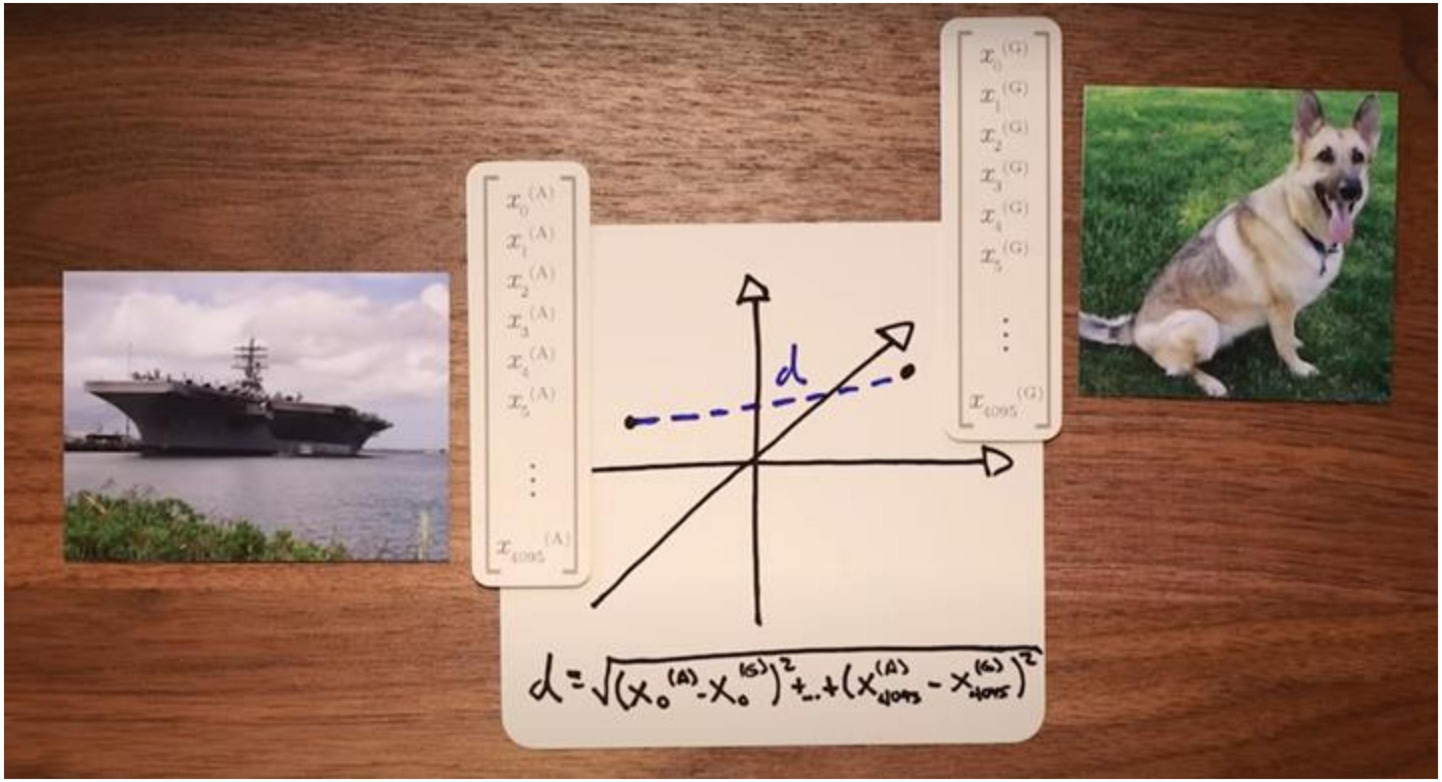
Second to last forms an embedding of the input image



https://www.youtube.com/watch?v=UZDiGooFs54

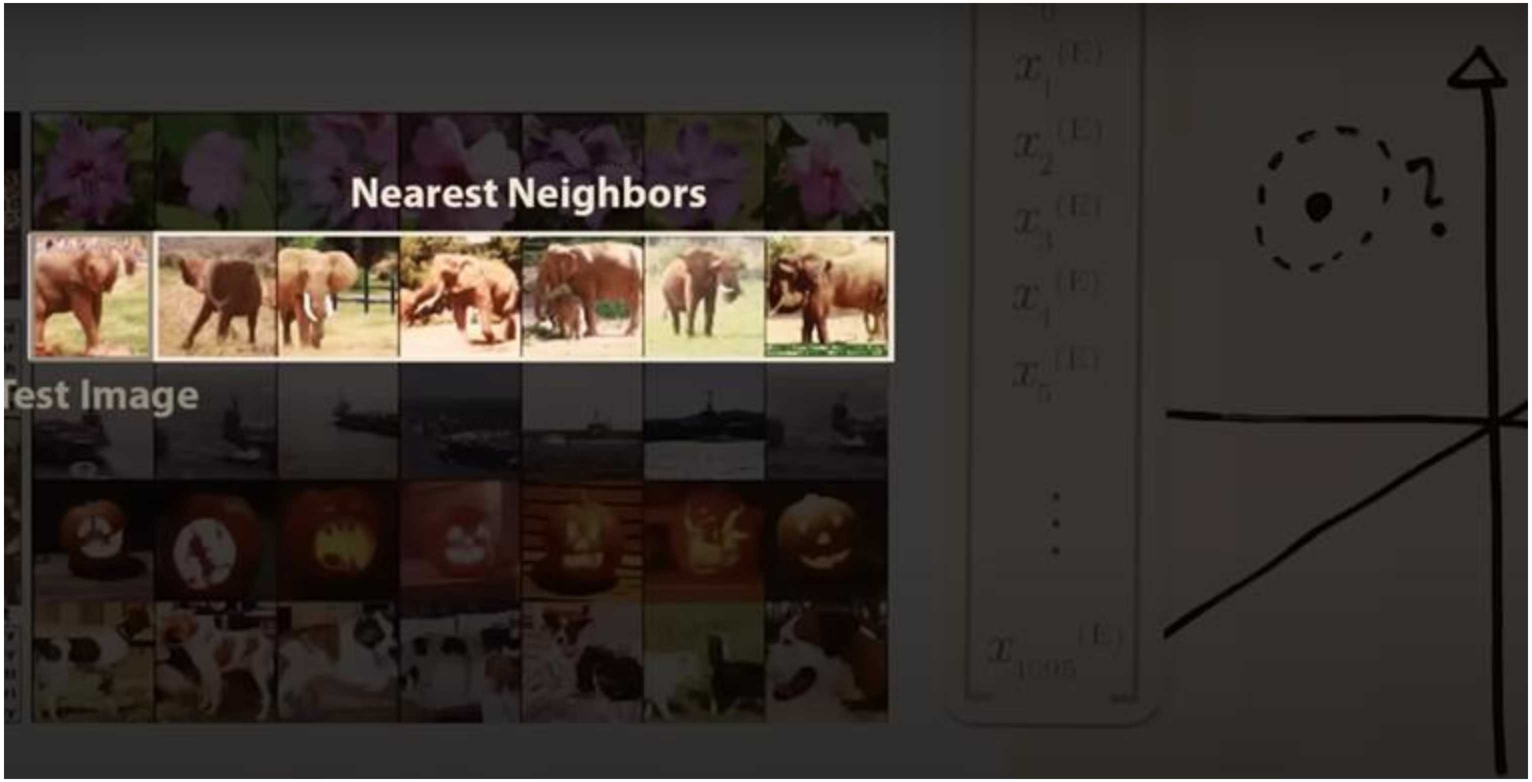
Each image creates a unique vector

Compute the distance between vectors in 4096-dim space



https://www.youtube.com/watch?v=UZDiGooFs54

Nearest neighbors in 4096-dim space are semantically related!
Robust to object orientation, size, color, etc

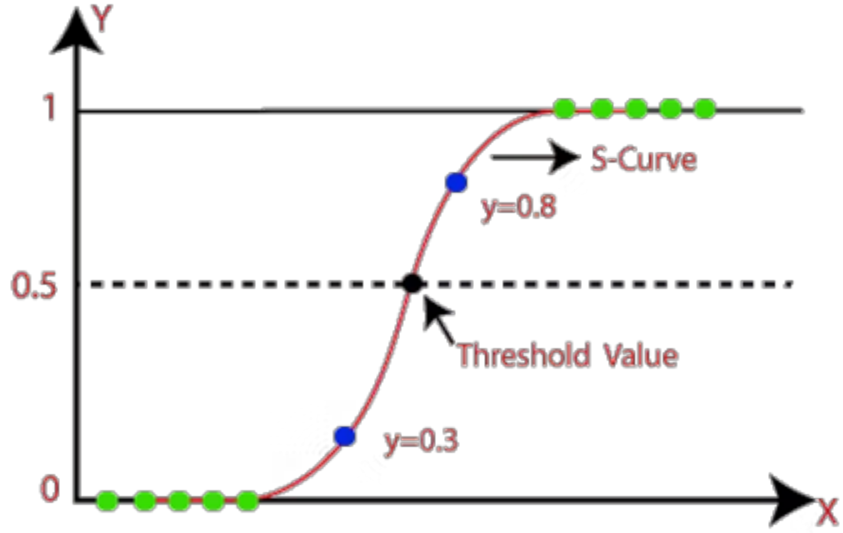


https://www.youtube.com/watch?v=UZDiGooFs54

Logistic Regression

Logistic regression assigns a weight to each feature to form a prediction

$$\hat{y}(w, x) = w_0 + w_1 x_1 + \ldots + w_p x_p$$



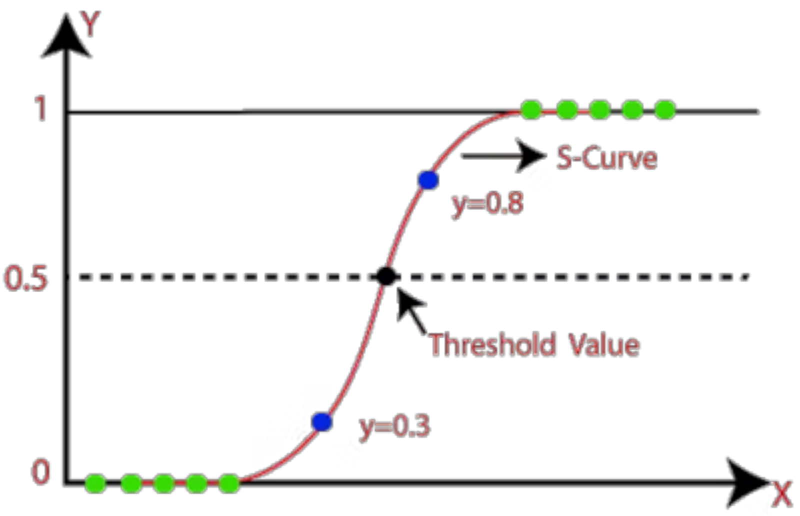
https://encord.com/blog/what-is-logistic-regression/

Logistic Regression

Logistic regression assigns a weight to each feature to form a prediction

Pros	Cons
• Easy to understand • Good accuracy when data is linearly separable • Less inclined to **overfit**	• Assumes linearity between output and features • No multicollinearity between features

$$\hat{y}(w, x) = w_0 + w_1 x_1 + \ldots + w_p x_p$$



https://encord.com/blog/what-is-logistic-regression/

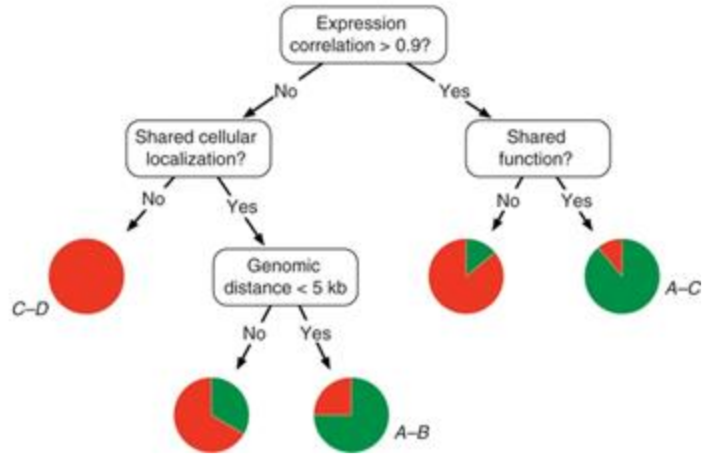
Decision Trees

Decision tree learns simple decision rules inferred from features

a

Gene Pair	Interact?	Expression correlation	Shared localization?	Shared function?	Genomic distance
A-B	Yes	0.77	Yes	No	1 kb
A-C	Yes	0.91	Yes	Yes	10 kb
C-D	No	0.1	No	No	1 Mb
⋮					

b

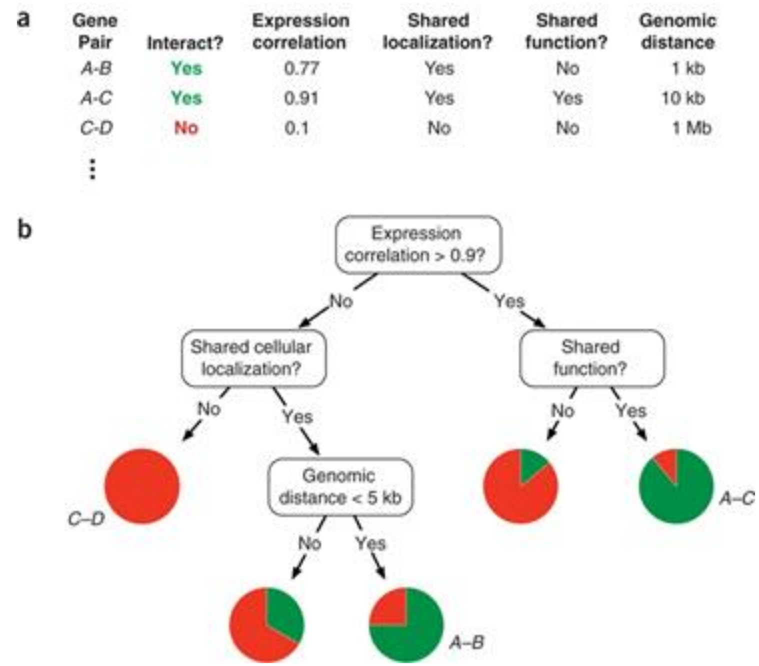


Kingsford, C., Salzberg, S. What are decision trees?. *Nat Biotechnol* 26, 1011–1013 (2008). https://doi.org/10.1038/nbt0908-1011

Decision Trees

Decision tree learns simple decision rules inferred from features

Pros	Cons
• Easy to understand • Can be visualized • Can handle multiple outputs	• Can create overcomplicated trees that overfit • Small variations in data can lead to completely different tree

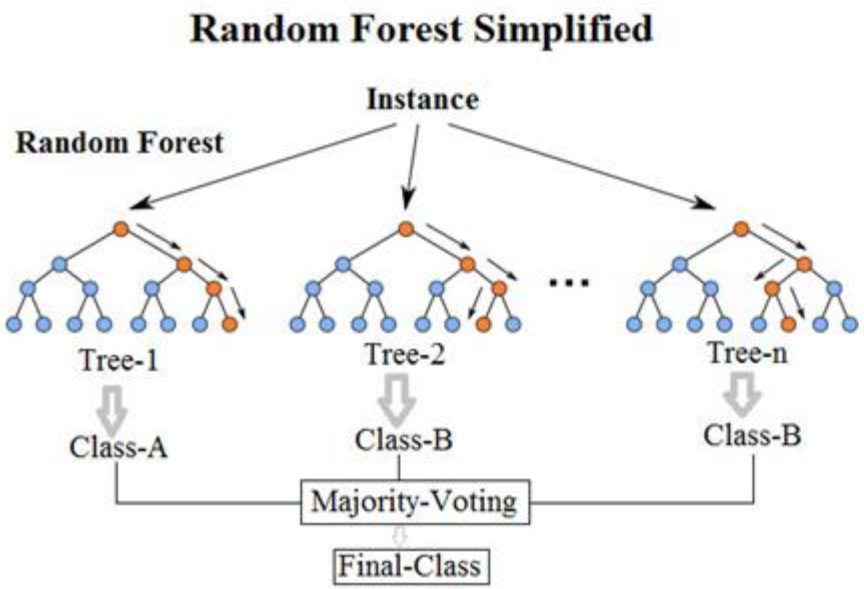


Kingsford, C., Salzberg, S. What are decision trees?. *Nat Biotechnol* 26, 1011–1013 (2008). https://doi.org/10.1038/nbt0908-1011

Random Forests

An ensemble of decision trees built from sample of training data

- Prediction is the averaged output of all decision trees



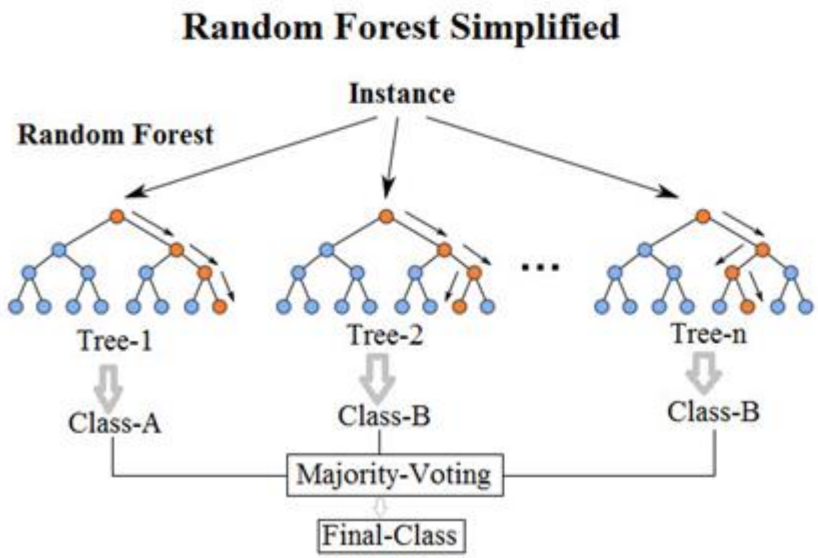
https://www.linkedin.com/pulse/random-forest-algorithm-interactive-discussion-niraj-kumar/

Random Forests

An ensemble of decision trees built from sample of training data

- Prediction is the averaged output of all decision trees

Pros	Cons
• Highly accurate • Intrinsically interpretable • Can learn non-linear decision boundaries • Parallel processing	• High computational complexity • Can overfit because of noise

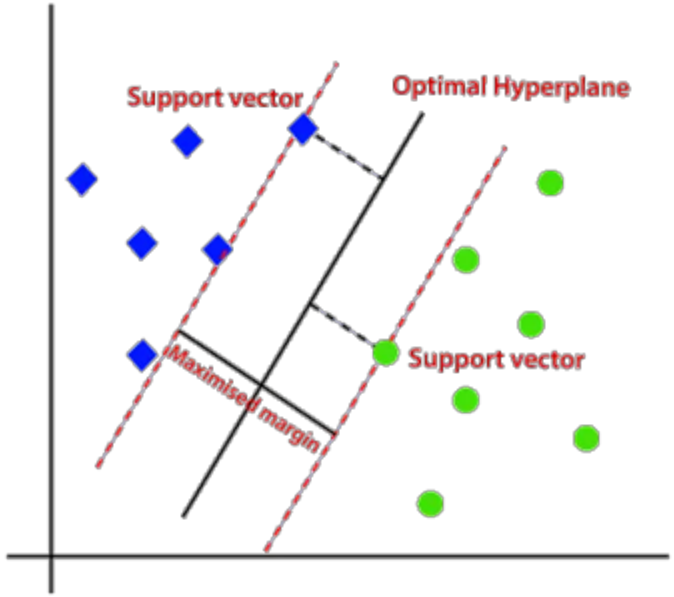


https://www.linkedin.com/pulse/random-forest-algorithm-interactive-discussion-niraj-kumar/

Support Vector Machine (SVM)

SVMs find a hyperplane that best separates the data into classes

- Maximize margin between different classes



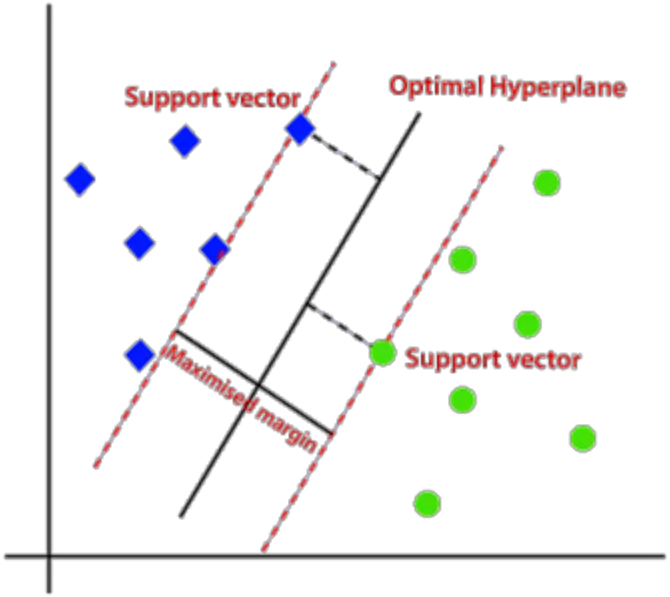
https://medium.com/@ambika199820/support-vector-machine-svm-algorithm-explained-c8facc10bcfa

SVM

Support Vector Machines find a hyperplane that best separates the data into classes

- Maximize margin between different classes

Pros	Cons
• Effectively handles high dimensional data • Can work well with small datasets • Can model non-linear decision boundaries	• Does not work for large data sets • Need to choose kernel • Limited to two-class problems • No probabilistic interpretation



https://medium.com/@ambika199820/support-vector-machine-svm-algorithm-explained-c8facc10bcfa

Reflections

- **CNNs are incredibly powerful for image recognition (and similar tasks)**
  - Heavily inspired by how the human brain processes visual information from the real world
  - Start with the most basic feature kernels and then iterate into higher level concepts leveraging their spatial proximity
  - Ultimately projects images into abstract semantic embeddings

- **Certain problems are a great fit for CNNs:**
  - ImageNet: Lots of test data, nicely annotated into 1000 categories
  - DeepVariant/Clairvoyante: Lots of test data; One network architecture made it the worlds best variant caller for all types of long read data: PacBio, ONT, 10x

- **Deeper network architectures generally perform better**
  - Largely constrained by GPU RAM & Cores

“Images” in Bioinformatics

- **DNA sequence: one-hot encode**
  - Are there other encoding schemes? (modDotPlot)
  - Predict features (genes, repeat classes); Predict species (kraken)
- **Coverage tracks:**
  - Predict: SNVs (DeepVariant), CNVs, SVs (Cue)
  - RNAseq, ChIPseq (overcome error, allele specific)
- **Functional Genomics**
  - Inputs
    - Expression from multiple tissues (GTEx) from one patient
    - Multiomics: Variants + expression + methylation
  - Predict phenotype: Gender (easy), cell types (easy), age (pretty easy)
    - Phenopredict: https://academic.oup.com/nar/article/46/9/e54/4920847
  - Disease outcome: Needs the right training data, right tissues
- **Multiple alignments, position weight matrix**
  - Predict conserved regions, predict binding sites, etc

	LeNet-5	AlexNet	GPT-3	GPT-4
YEAR	1998	2012	2020	2023
TRAINING DATA	· MNIST Dataset · 60k training examples · 10 classes	· ImageNet Dataset (ILSVRC) · 1.2M training examples · 1000 classes	· Common Crawl, WebText, Wikipedia, others · ~500B training tokens · ~100k unique tokens	· ~13T training tokens*
TRAINING COMPUTE	· Pentium II CPU · ~0.27 GFLOPs	· Dual Nvidia GTX 580 · 3162 GFLOPs	· 10,000 Nvidia V100 GPUs · 1+ ExaFLOPs	· 25,000 Nvidia A100s GPUs* · ~4+ ExaFLOPs
ALGORITHM	· ~60k Parameters · 5 Layers · Sigmoid Activation Function	· ~60M Parameters · 8 Layers · ReLU Activation Function · Dropout	· 175B Parameters · 96 Layers · Transformers	· 1T+ Parameters* · *120 Layers · Transformers

https://www.youtube.com/watch?v=UZDiGooFs54

Functional Genomics

14	10/13/25	M	Functional Analysis 1: RNA-seq		* RNA-Seq: a revolutionary tool for transcriptomics (Wang et al, 2009. Nature Reviews Genetics) * Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks (Trapnell et al, 2012, Nature Protocols) * Salmon provides fast and bias-aware quantification of transcript expression (Patro et al, 2017, Nature Methods) * Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications (Krueger and Andrews, 2011, Bioinformatics)
15	10/15/25	W	Functional Analysis 2: Single Cell Genomics	Ass4	* Ginkgo: Interactive analysis and assessment of single-cell copy-number variations (Garvin et al, 2015, Nature Methods) * The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells (Trapnell et al, Nature Biotech, 2014) * Eleven grand challenges in single-cell data science (Lahnemann et al, Genome Biology, 2020)
16	10/20/25	M	Functional Analysis 3: Ab initiio gene finding		* BLAST: Basic Local Alignment Search Tool * Glimmer: Microbial gene identification using interpolated Markov models * MAKER2: an annotation pipeline and genome-database management tool for second-generation genome projects * BEDTools: a flexible suite of utilities for comparing genomic features (Quinlan & Hall, 2010, Bioinformatics)
17	10/22/25	W	Functional Analysis 4: Methyl-seq, Chip-seq, and Hi-C	Prelim report assigned	* ChIP-seq and beyond: new and improved methodologies to detect and characterize protein-DNA interactions (Furey, 2012, Nature Reviews Genetics) * PeakSeq enables systematic scoring of ChIP-seq experiments relative to controls (Rozowsky et al. 2009. Nature Biotech) * Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome (Lieberman-Aiden et al, 2009, Science)
18	10/27/25	M	Functional Analysis 5: Regulatory States, ENCODE, GTEx, RoadMap		* An integrated encyclopedia of DNA elements in the human genome (The ENCODE Project Consortium, Nature, 2012) * Genetic effects on gene expression across human tissues (GTEx Consortium, Nature, 2017) * Integrative analysis of 111 reference human epigenomes (Roadmap Epigenome Consortium, Nature, 2015) * ChromHMM: automating chromatin-state discovery and characterization (Ernst & Kellis, 2012, Nature Methods) * Segway: Unsupervised pattern discovery in human chromatin structure through genomic segmentation (Hoffman et al, 2012, Nature Methods)
19	10/29/25	W	Transformers	Ass5	* Attention is all you need (Vaswani et al. 2017, arXiv)
20	11/3/25	M	Enformer + Other DL applications		* Effective gene expression prediction from sequence by integrating long-range interactions (Avsec et al., 2021, Nature Methods) * Personal transcriptome variation is poorly explained by current genomic deep learning models (Huang et al., 2023, Nature Genetics) * Benchmarking of deep neural networks for predicting personal gene expression from DNA sequence highlights shortcomings (Sasse et al., 2023, Nature Genetics)
21	11/5/25	W	Project support day		
22	11/10/25	M	Review 2	Final Report Assigned	
23	11/12/25	W	Exam 2 (In class)		
24	11/17/25	M	Wrap up		* Deep Learning Sequence Models for Transcriptional Regulation (Sokolova et al., 2024, Annual Reviews of Genomics and Human Genetics) * AlphaFold (Jumper et al, 2021, Nature)

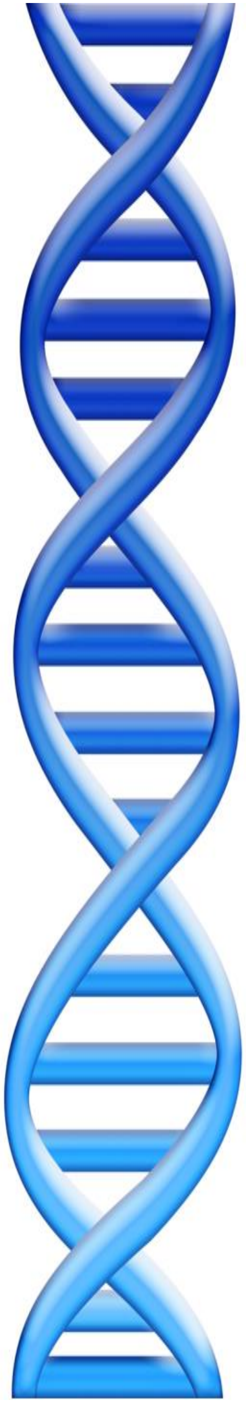
Goal: Genome Annotations

aatgcatgcggctatgctaatgcatgcggctatgctaagctgggatccgatgacaatgcatgcggctatgctaat
gcatgcggctatgcaagctgggatccgatgactatgctaagctgggatccgatgacaatgcatgcggctatgct
aatgaatggtcttgggatttaccttggaatgctaagctgggatccgatgacaatgcatgcggctatgctaatgaa
tggtcttgggatttaccttggaatatgctaatgcatgcggctatgctaagctgggatccgatgacaatgcatgcg
gctatgctaatgcatgcggctatgcaagctgggatccgatgactatgctaagctgcggctatgctaatgcatgcg
gctatgctaagctgggatccgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctgggatcct
gcggctatgctaatgaatggtcttgggatttaccttggaatgctaagctgggatccgatgacaatgcatgcggct
atgctaatgaatggtcttgggatttaccttggaatatgctaatgcatgcggctatgctaagctgggaatgcatgcg
gctatgctaagctgggatccgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctgggatccg
atgactatgctaagctgcggctatgctaatgcatgcggctatgctaagctcatgcggctatgctaagctgggaat
gcatgcggctatgctaagctgggatccgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctg
ggatccgatgactatgctaagctgcggctatgctaatgcatgcggctatgctaagctcggctatgctaatgaatg
gtcttgggatttaccttggaatgctaagctgggatccgatgacaatgcatgcggctatgctaatgaatggtcttgg
gatttaccttggaatatgctaatgcatgcggctatgctaagctgggaatgcatgcggctatgctaagctgggatc
cgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctgggatccgatgactatgctaagctgcg
gctatgctaatgcatgcggctatgctaagctcatgcggctatgctaagctcatgcgg

Goal: Genome Annotations

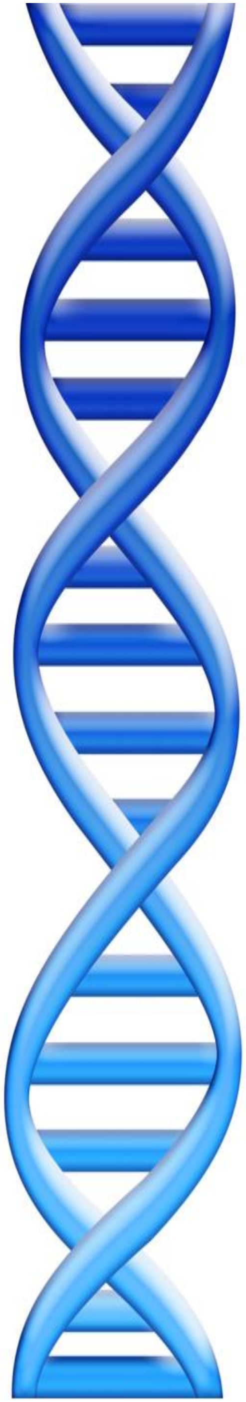
aatgcatgcggctatgctaatgcatgcggctatgctaagctgggatccgatgacaatgcatgcggctatgctaat
gcatgcggctatgcaagctgggatccgatgactatgctaagctgggatccgatgacaatgcatgcggctatgct
aatgaatggtcttgggatttaccttggaatgctaagctgggatccgatgacaatgcatgcggctatgctaatgaa
tggtcttgggatttaccttggaatatgctaatgcatgcggctatgctaagctgggatccgatgacaatgcatgcg
gctatgctaatgcatgcggctatgcaagctgggatccgatgactatgctaagctgcggctatgctaatgcatgcg
gctatgctaagctgggatccgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctgggatcct
gcggctatgctaatgaatggtctt ... ggatccgatgacaatgcatgcggct
atgctaatgaatggtcttgggatt ... gctatgctaagctgggaatgcatgcg
gctatgctaagctgggatccgat ... atgcggctatgcaagctgggatccg
atgactatgctaagctgcggctatgctaatgcatgcggctatgctaagctcatgcggctatgctaagctgggaat
gcatgcggctatgctaagctgggatccgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctg
ggatccgatgactatgctaagctgcggctatgctaatgcatgcggctatgctaagctcggctatgctaatgaatg
gtcttgggatttaccttggaatgctaagctgggatccgatgacaatgcatgcggctatgctaatgaatggtcttgg
gatttaccttggaatatgctaatgcatgcggctatgctaagctgggaatgcatgcggctatgctaagctgggatc
cgatgacaatgcatgcggctatgctaatgcatgcggctatgcaagctgggatccgatgactatgctaagctgcg
gctatgctaatgcatgcggctatgctaagctcatgcggctatgctaagctcatgcgg

Gene!



Outline

1. Alignment to other genomes
2. Prediction aka “Gene Finding”
3. Experimental & Functional Assays



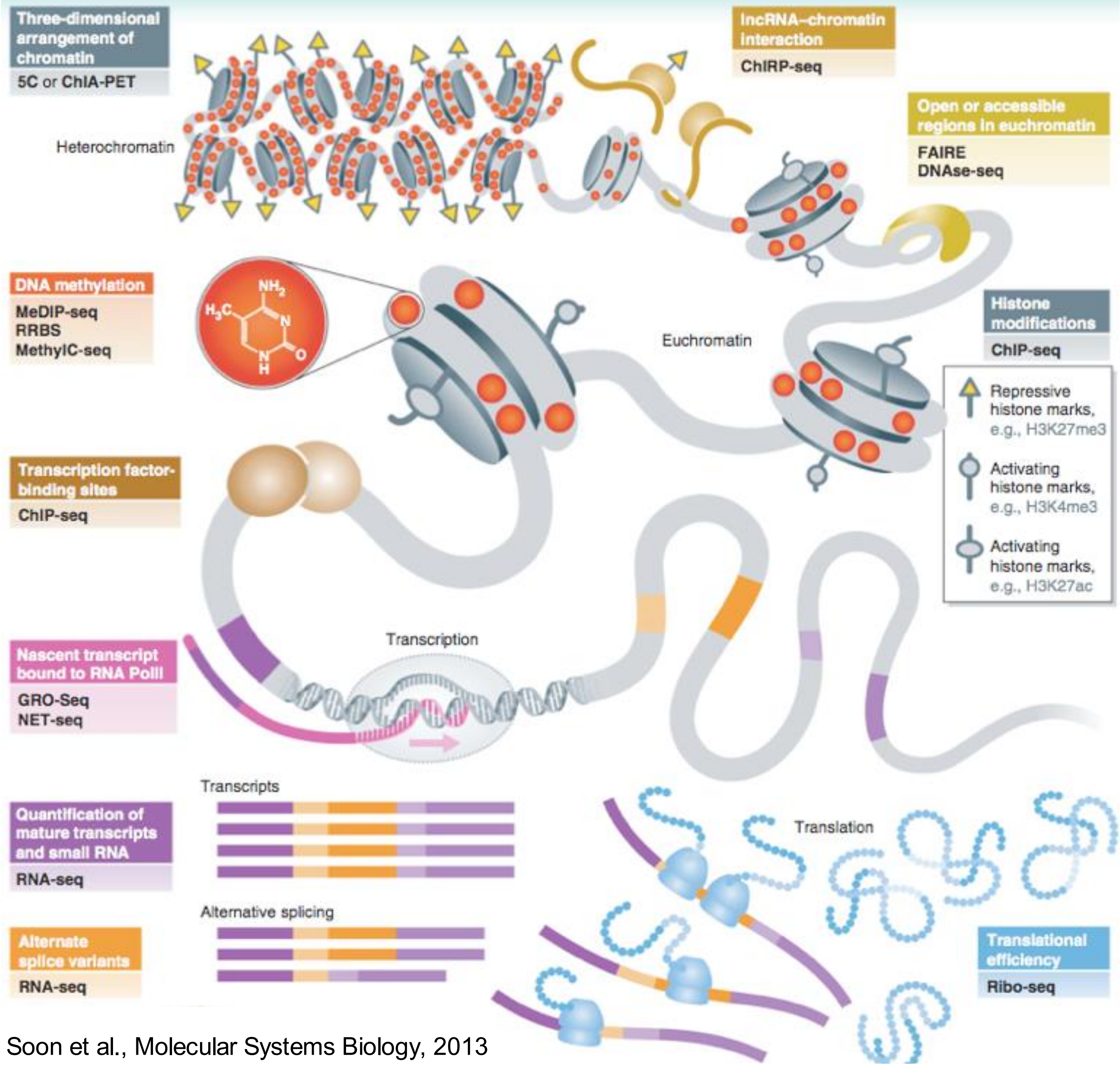
Outline

1. Alignment to other genomes
2. Prediction aka "Gene Finding"
3. Experimental & Functional Assays

Sequencing Assays

The *Seq List (in chronological order)

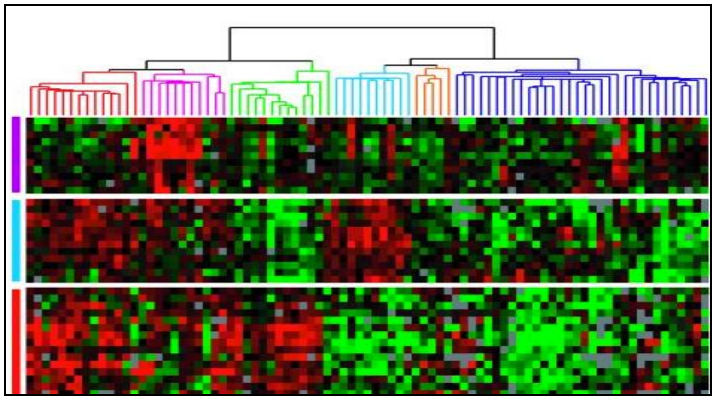
1. Gregory E. Crawford et al., “Genome-wide Mapping of DNase Hypersensitive Sites Using Massively Parallel Signature Sequencing (MPSS),” Genome Research 16, no. 1 (January 1, 2006): 123–131, doi:10.1101/gr.4074106.
2. David S. Johnson et al., “Genome-Wide Mapping of in Vivo Protein-DNA Interactions,” Science 316, no. 5830 (June 8, 2007): 1497–1502, doi:10.1126/science.1141319.
3. Tarjei S. Mikkelsen et al., “Genome-wide Maps of Chromatin State in Pluripotent and Lineage-committed Cells,” Nature 448, no. 7153 (August 2, 2007): 553–560, doi:10.1038/nature06008.
4. Thomas A. Down et al., “A Bayesian Deconvolution Strategy for Immunoprecipitation-based DNA Methylome Analysis,” Nature Biotechnology 26, no. 7 (July 2008): 779–785, doi:10.1038/nbt1414.
5. Ali Mortazavi et al., “Mapping and Quantifying Mammalian Transcriptomes by RNA-Seq,” Nature Methods 5, no. 7 (July 2008): 621–628, doi:10.1038/nmeth.1226.
6. Nathan A. Baird et al., “Rapid SNP Discovery and Genetic Mapping Using Sequenced RAD Markers,” PLoS ONE 3, no. 10 (October 13, 2008): e3376, doi:10.1371/journal.pone.0003376.
7. Leighton J. Core, Joshua J. Waterfall, and John T. Lis, “Nascent RNA Sequencing Reveals Widespread Pausing and Divergent Initiation at Human Promoters,” Science 322, no. 5909 (December 19, 2008): 1845–1848, doi:10.1126/science.1162228.
8. Chao Xie and Martti T. Tammi, “CNV-seq, a New Method to Detect Copy Number Variation Using High-throughput Sequencing,” BMC Bioinformatics 10, no. 1 (March 6, 2009): 80, doi:10.1186/1471-2105-10-80.
9. Jay R. Hesselberth et al., “Global Mapping of protein-DNA Interactions in Vivo by Digital Genomic Footprinting,” Nature Methods 6, no. 4 (April 2009): 283–289, doi:10.1038/nmeth.1313.
10. Nicholas T. Ingolia et al., “Genome-Wide Analysis in Vivo of Translation with Nucleotide Resolution Using Ribosome Profiling,” Science 324, no. 5924 (April 10, 2009): 218–223, doi:10.1126/science.1168978.
11. Alayne L. Brunner et al., “Distinct DNA Methylation Patterns Characterize Differentiated Human Embryonic Stem Cells and Developing Human Fetal Liver,” Genome Research 19, no. 6 (June 1, 2009): 1044–1056, doi:10.1101/gr.088773.108.
12. Mayumi Oda et al., “High-resolution Genome-wide Cytosine Methylation Profiling with Simultaneous Copy Number Analysis and Optimization for Limited Cell Numbers,” Nucleic Acids Research 37, no. 12 (July 1, 2009): 3829–3839, doi:10.1093/nar/gkp260.
13. Zachary D. Smith et al., “High-throughput Bisulfite Sequencing in Mammalian Genomes,” Methods 48, no. 3 (July 2009): 226–232, doi:10.1016/j.ymeth.2009.05.003.
14. Andrew M. Smith et al., “Quantitative Phenotyping via Deep Barcode Sequencing,” Genome Research (July 21, 2009).



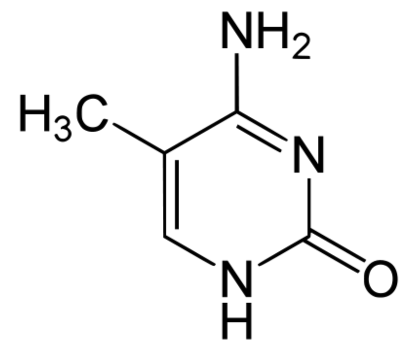
Soon et al., Molecular Systems Biology, 2013

*-seq in 4 short vignettes

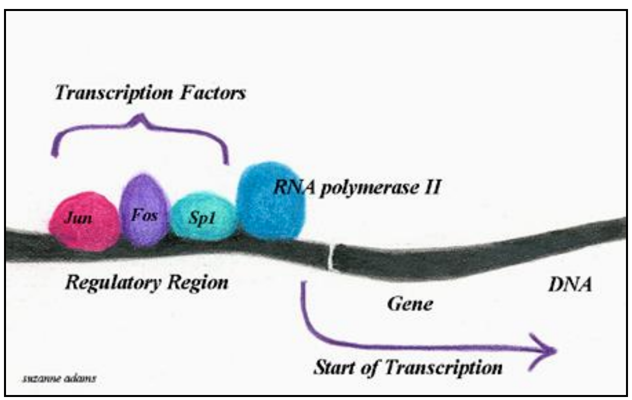
RNA-seq



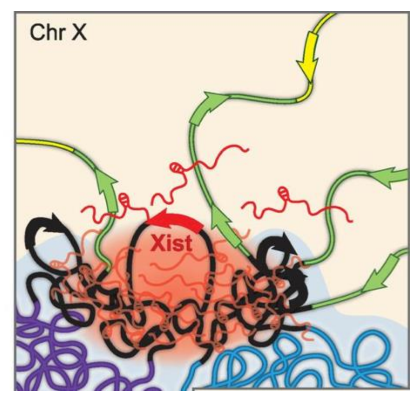
Methyl-seq



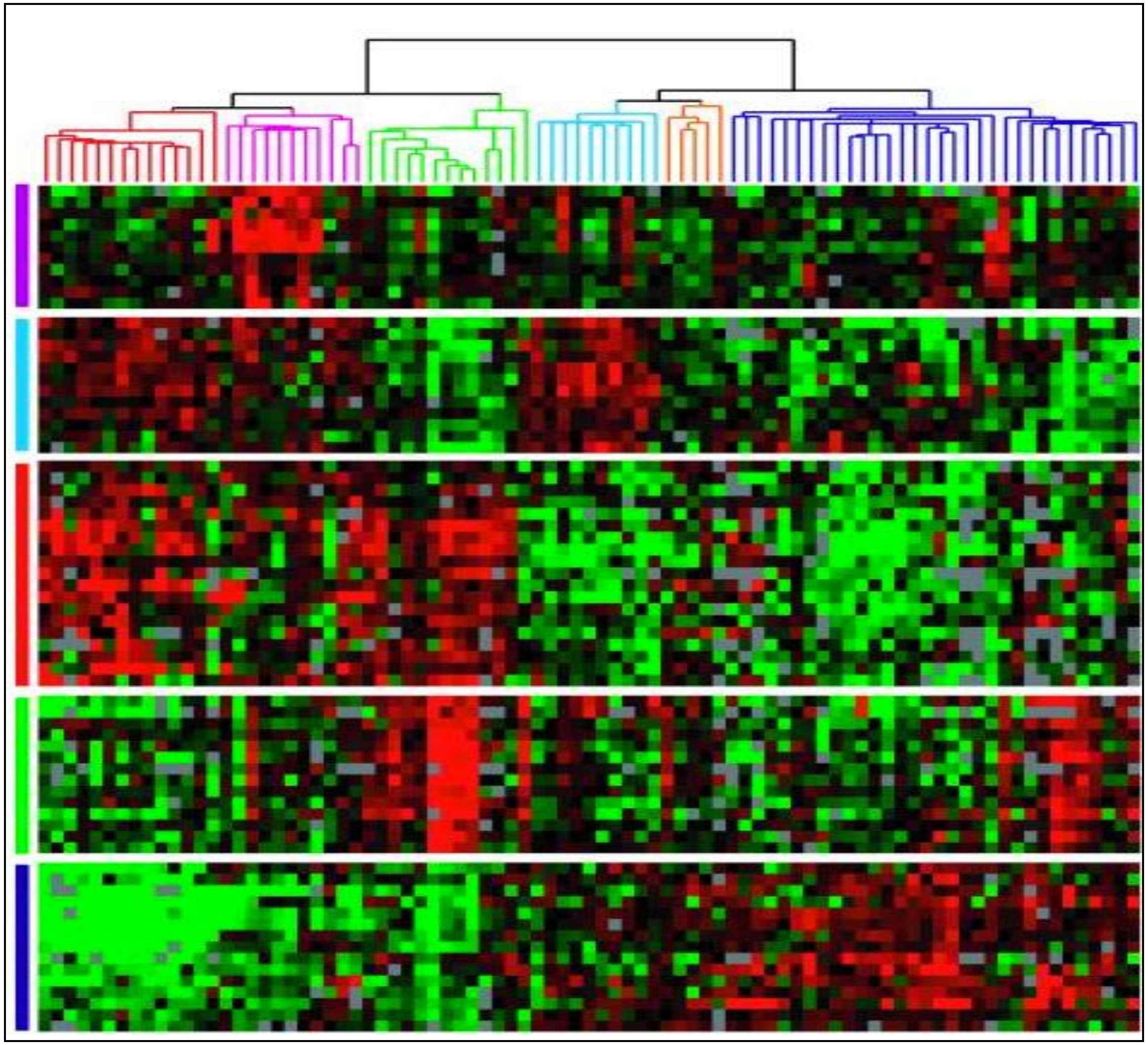
ChIP-seq



Hi-C

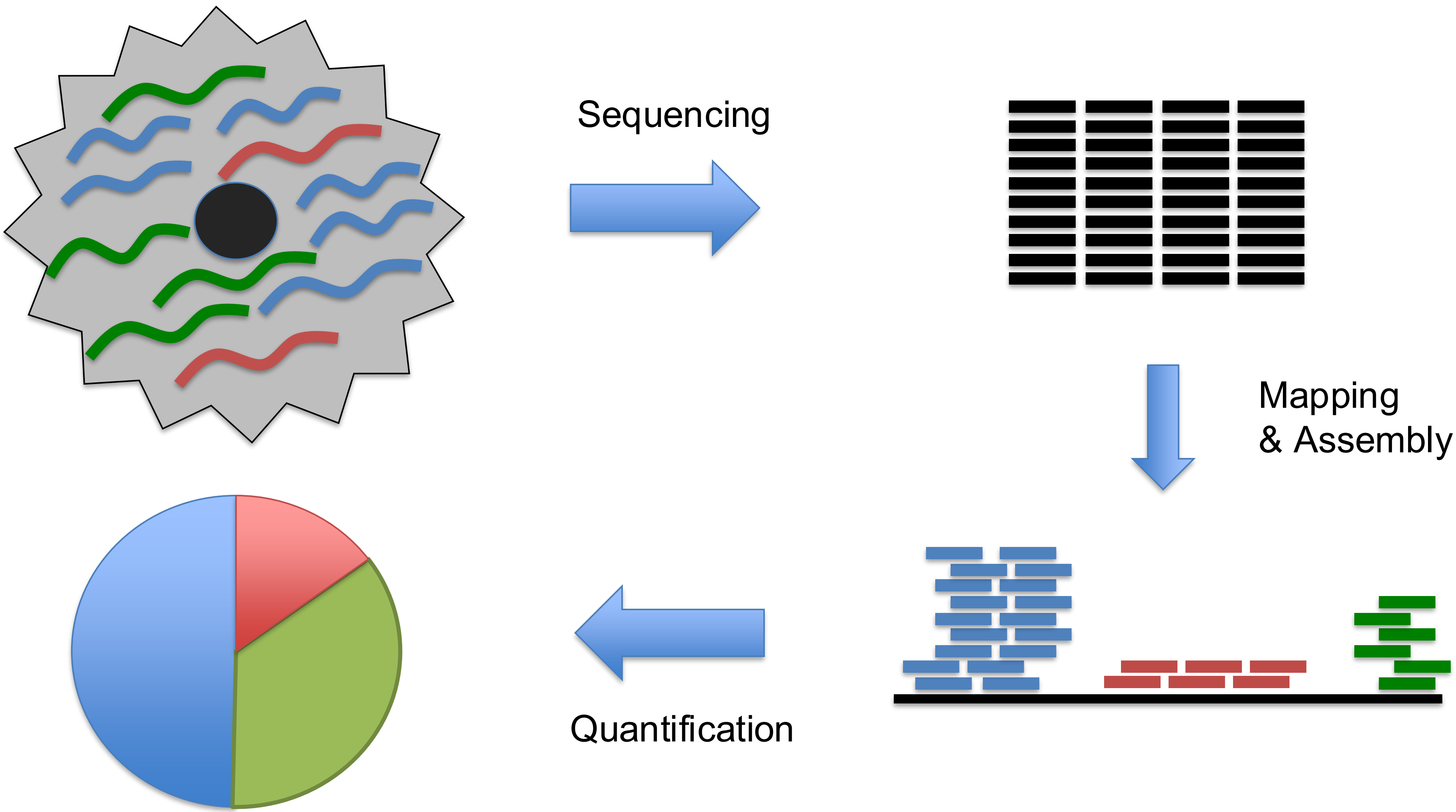


RNA-seq

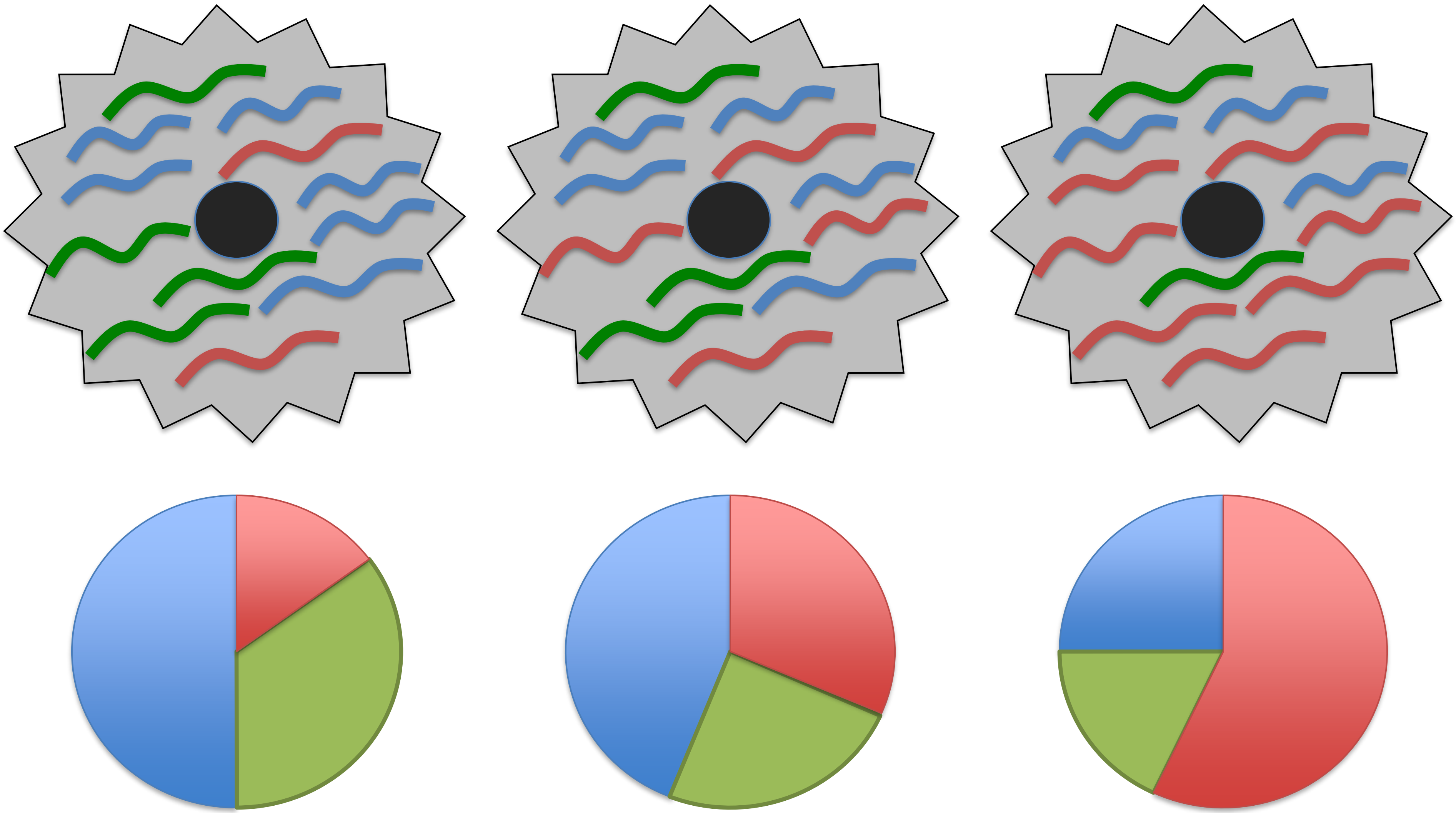


Gene expression patterns of breast carcinomas distinguish tumor subclasses with clinical implications.
Sørlie et al (2001) *PNAS.* 98(19):10869-74.

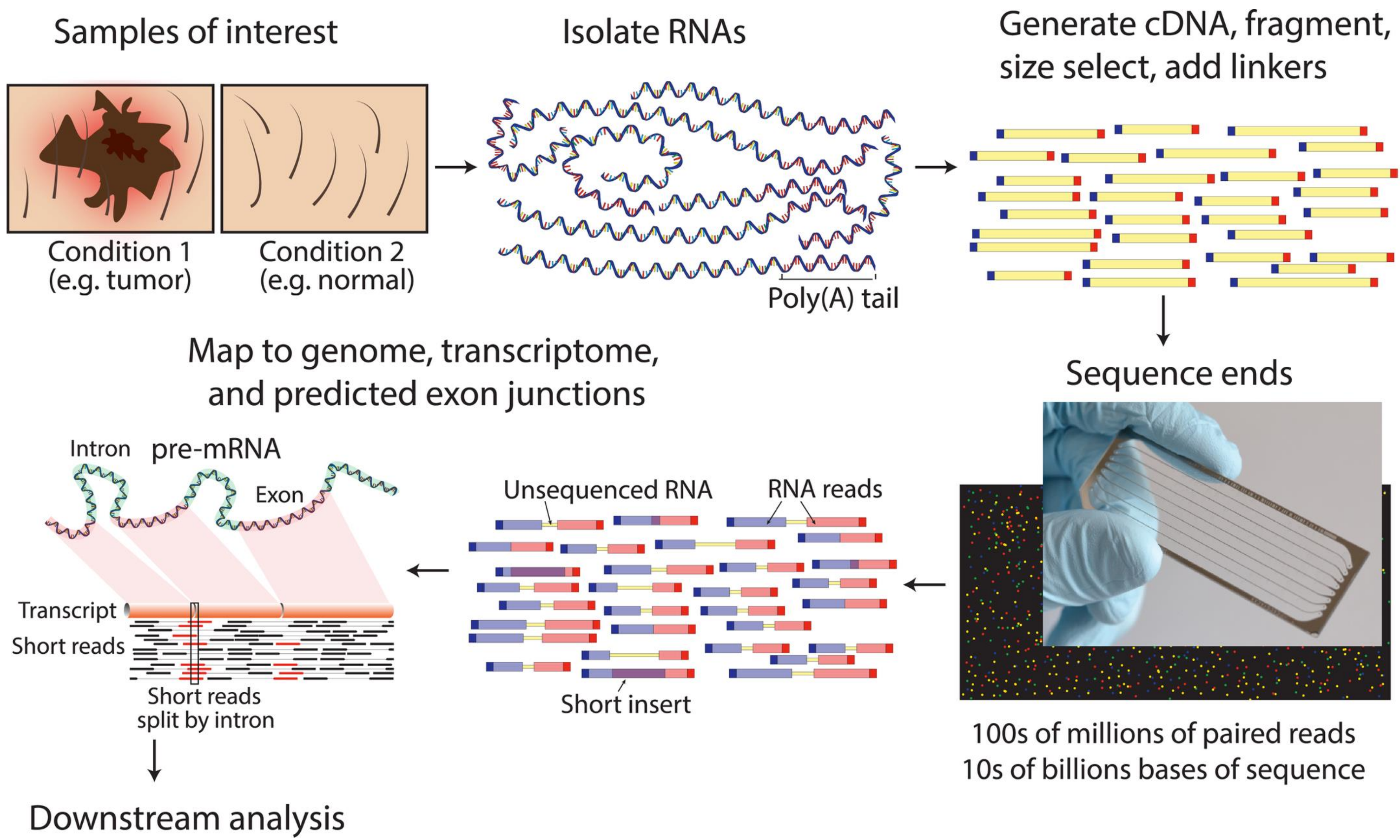
RNA-seq Overview



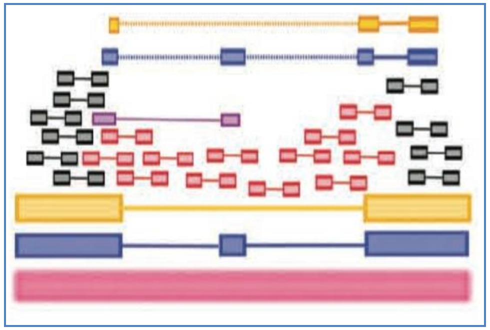
RNA-seq Overview



RNA-seq Overview



RNA-seq Challenges



Challenge 1: Eukaryotic genes are spliced

RNA-Seq Approaches

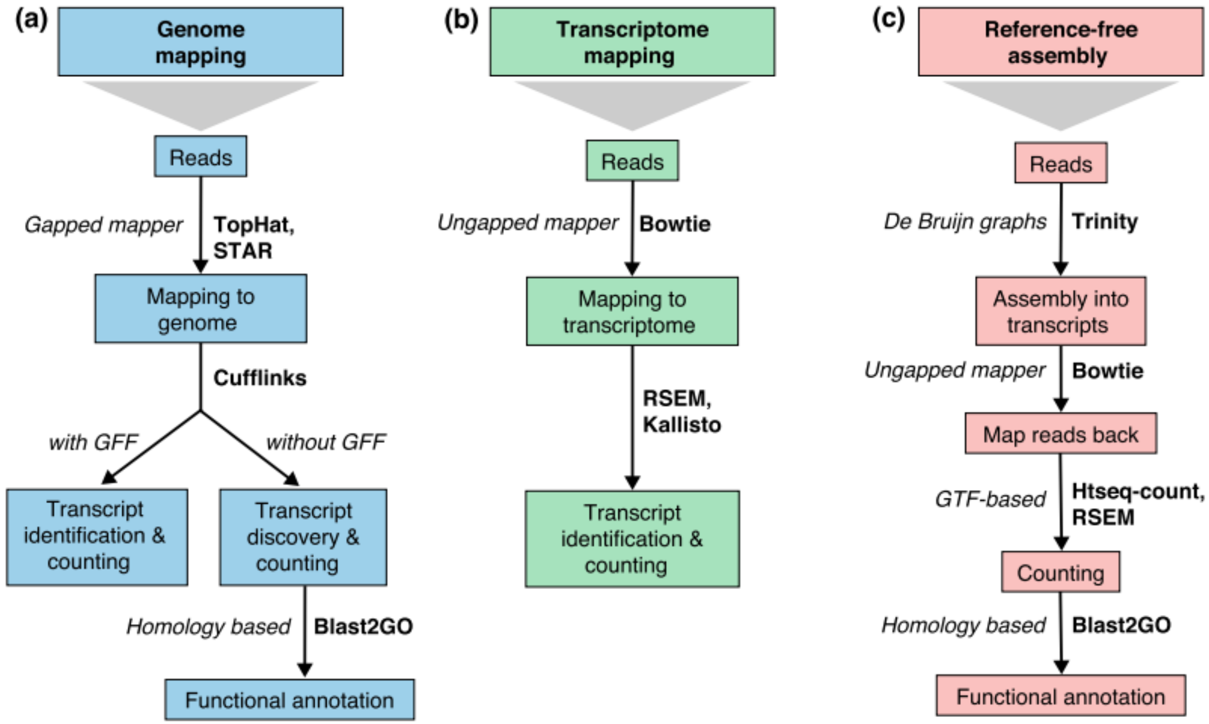


Fig. 2 Read mapping and transcript identification strategies. Three basic strategies for regular RNA-seq analysis. **a** An annotated genome is available and reads are mapped to the genome with a gapped mapper. Next (novel) transcript discovery and quantification can proceed with or without an annotation file. Novel transcripts are then functionally annotated. **b** If no novel transcript discovery is needed, reads can be mapped to the reference transcriptome using an ungapped aligner. Transcript identification and quantification can occur simultaneously. **c** When no genome is available, reads need to be assembled first into contigs or transcripts. For quantification, reads are mapped back to the novel reference transcriptome and further analysis proceeds as in (**b**) followed by the functional annotation of the novel transcripts as in (**a**). Representative software that can be used at each analysis step are indicated in *bold text*. Abbreviations: *GFF* General Feature Format, *GTF* gene transfer format, *RSEM* RNA-Seq by Expectation Maximization

A survey of best practices for RNA-seq data analysis
Conesa et al (2016) Genome Biology. doi 10.1186/s13059-016-0881-8

RNA-Seq Approaches

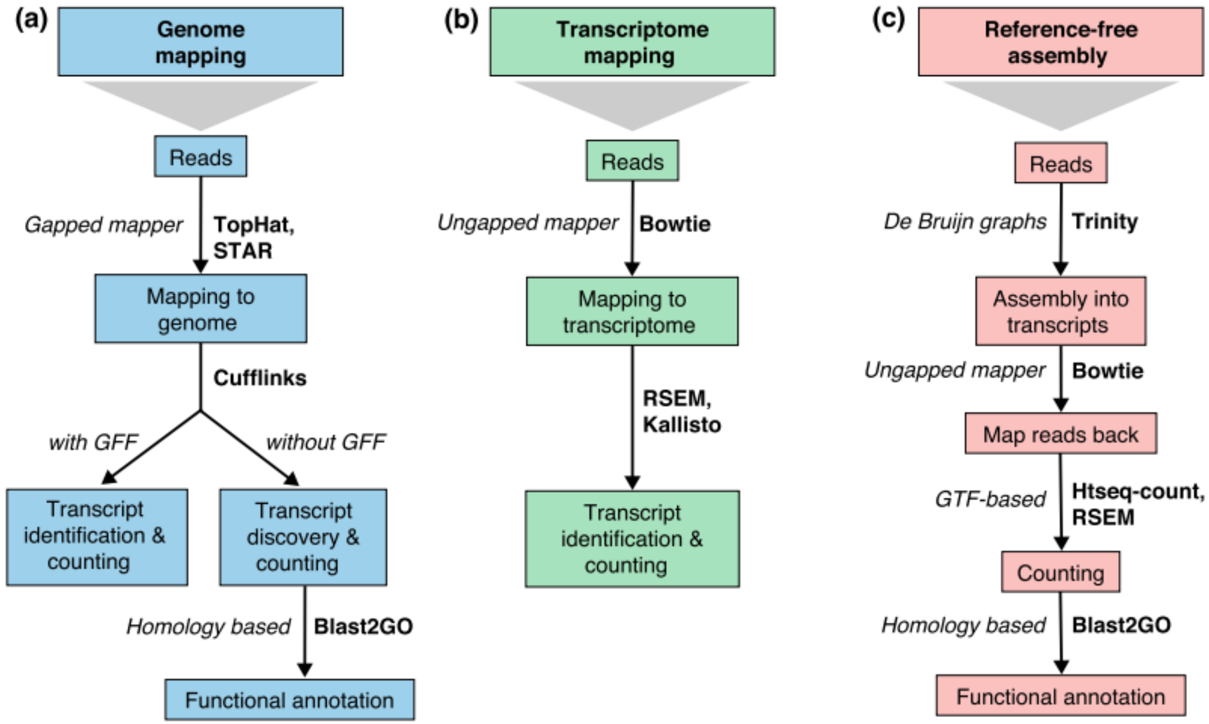


Fig. 2 Read mapping and transcript identification strategies. Three basic strategies for regular RNA-seq analysis. **a** An annotated genome is available and re … ovel) transcript discovery and quantification can proceed with or without an ann … no novel transcript discovery is needed, reads can be mapped to the reference transcriptome using an ungapped aligner. Transcript identification and quantification can occur simultaneously. **c** When no genome is available, reads n … o contigs or transcripts. For quantification, reads are mapped back to the novel reference transcriptome and further a … owed by the functional annotation of the novel transcripts as in (**a**). Representative software that can be used a … ated in *bold text*. Abbreviations: *GFF* General Feature Format, *GTF* gene transfer format, *RSEM* RNA-Seq by Expectation Maximization

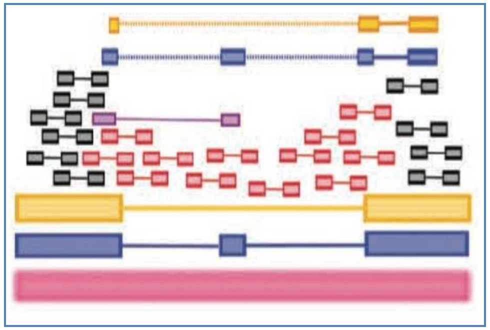
Which approach should we use?

It depends….

A survey of best practices for RNA-seq data analysis

Conesa et al (2016) Genome Biology. doi 10.1186/s13059-016-0881-8

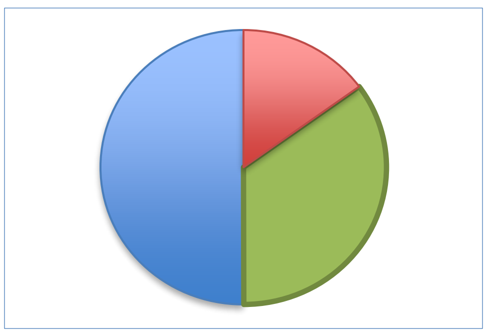
RNA-seq Challenges



Challenge 1: Eukaryotic genes are spliced

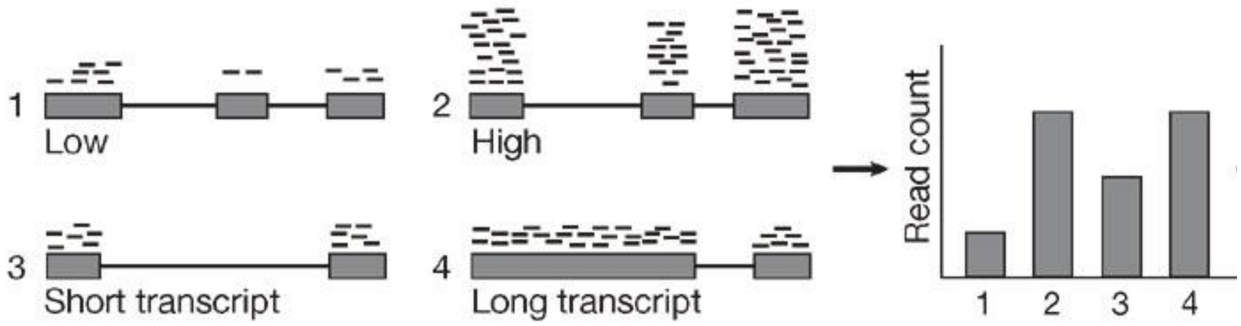
Solution: Use a spliced aligner, and assemble isoforms

TopHat: discovering spliced junctions with RNA-Seq.
Trapnell et al (2009) *Bioinformatics*. 25:0 1105-1111



Challenge 2: Read Count != Transcript abundance

RPKM, FPKM, TPM

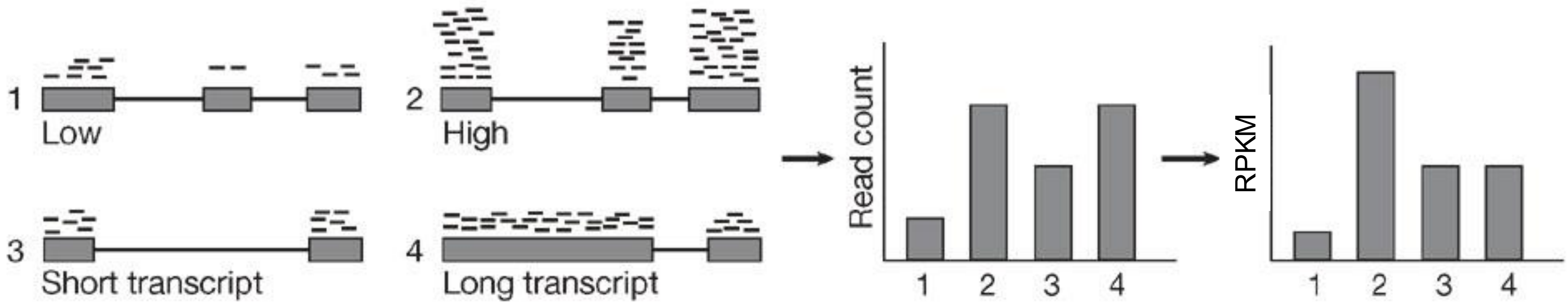


Counting Reads that align to a gene DOESN'T work!

- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)

RPKM, FPKM, TPM



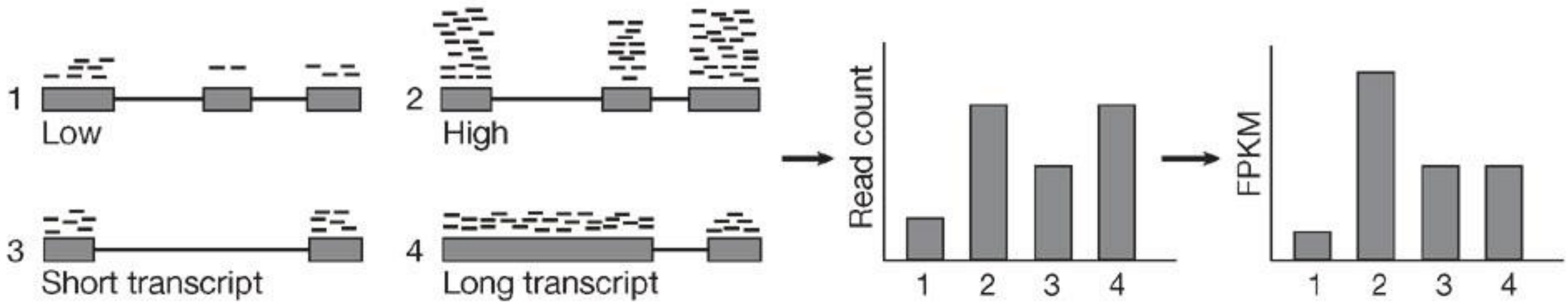
Counting Reads that align to a gene DOESN'T work!
- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)

(Count reads aligned to gene) / (length of gene in kilobases) / (# millions of read mapped)

=> Wait a second, reads in a pair arent independent!

RPKM, FPKM, TPM



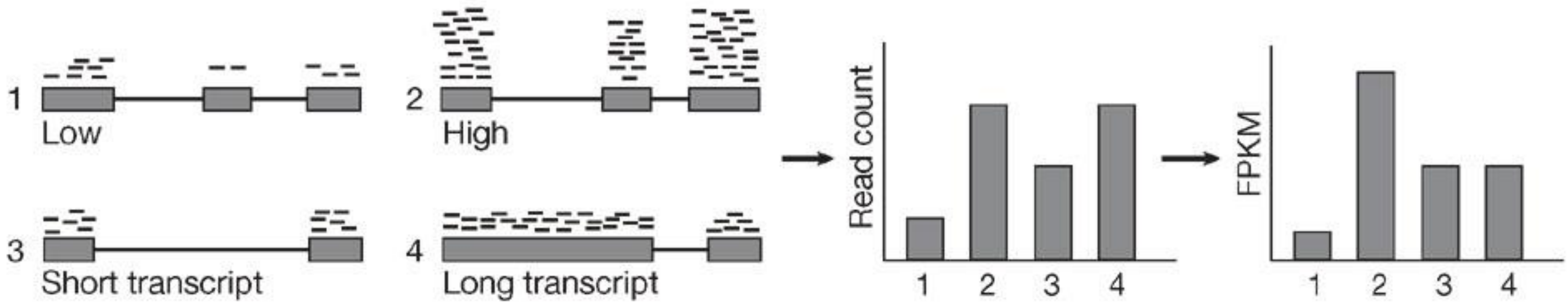
Counting Reads that align to a gene DOESN'T work!
- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)
=> Wait a second, reads in a pair arent independent!

2. FPKM: Fragments Per Kilobase of Exon Per Million Reads Mapped (Trapnell et al, 2010)
⇒ Does a much better job with short exons & short genes by boosting coverage

⇒ Wait a second, FPKM depends on the average transcript length!

RPKM, FPKM, TPM



Counting Reads that align to a gene DOESN'T work!
- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)
=> Wait a second, reads in a pair arent independent!

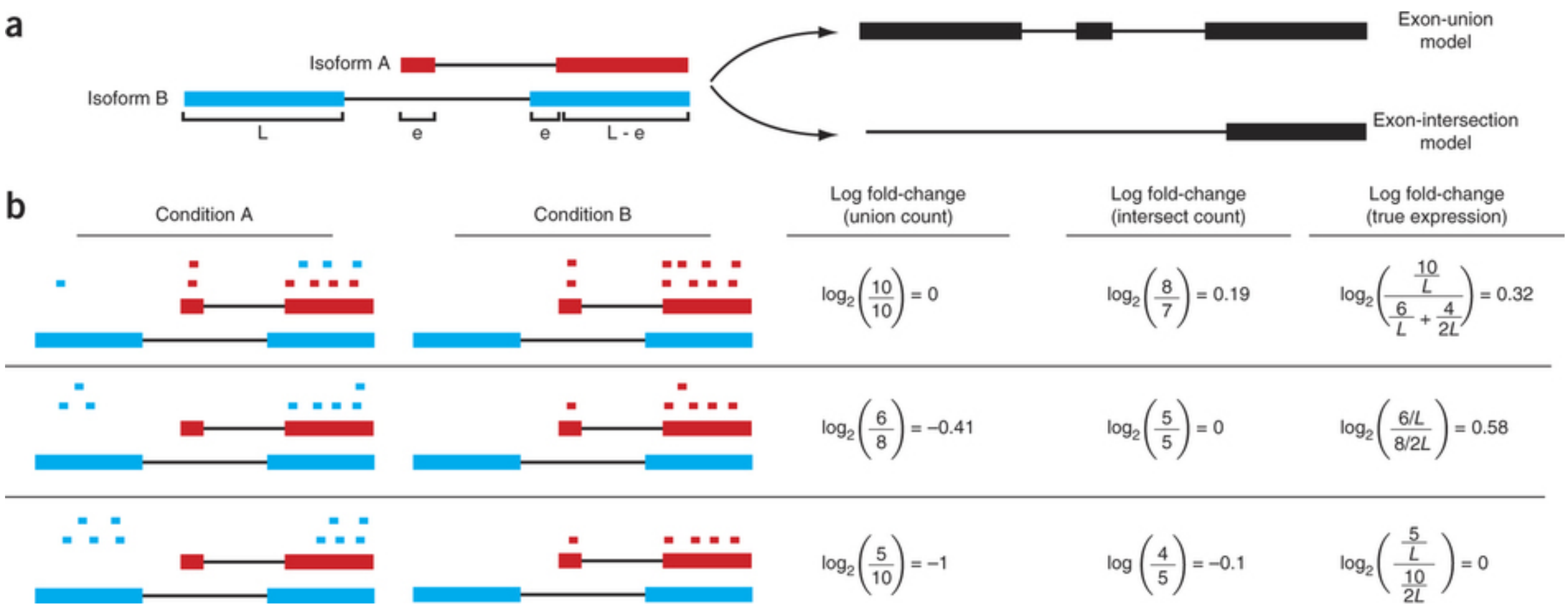
2. FPKM: Fragments Per Kilobase of Exon Per Million Reads Mapped (Trapnell et al, 2010)
=> Wait a second, FPKM depends on the average transcript length!

3. TPM: Transcripts Per Million (Li et al, 2011)
⇒ If you were to sequence one million full length transcripts, TPM is the number of transcripts you would have seen of type i, given the abundances of the other transcripts in your sample

=> Recommend you use TPM for all analysis, easy to compute given FPKM

$$\mathrm{TPM}_i = \left(\frac{\mathrm{FPKM}_i}{\sum_j \mathrm{FPKM}_j}\right) \cdot 10^6$$

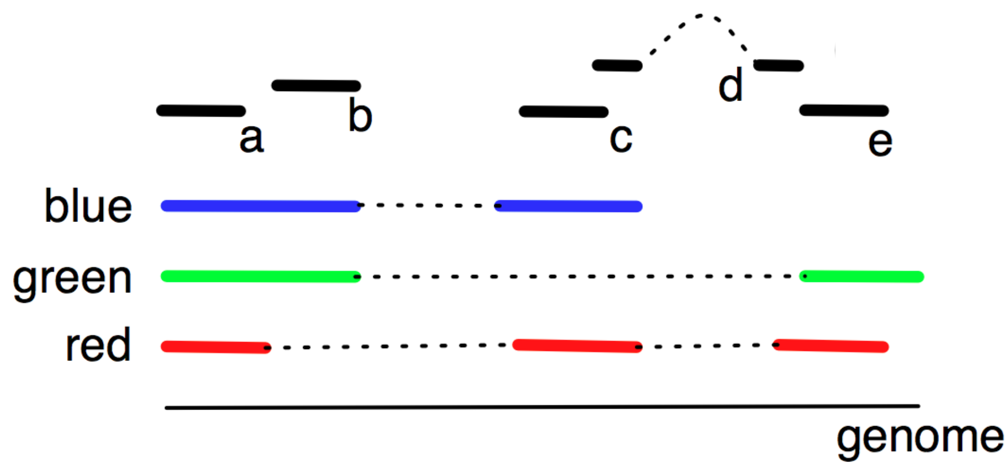
Gene or Isoform Quantification?



Key point : The length of the actual molecule from which the fragments derive is crucially important to obtaining accurate abundance estimates.

Differential analysis of gene regulation at transcript resolution with RNA-seq
Trapnell et al (2013) Nature Biotechnology 31, 46–53. doi:10.1038/nbt.2450

Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



The gene has three isoforms (red, green, blue) of the same length.
Our initial expectation is all 3 isoforms are equally expressed

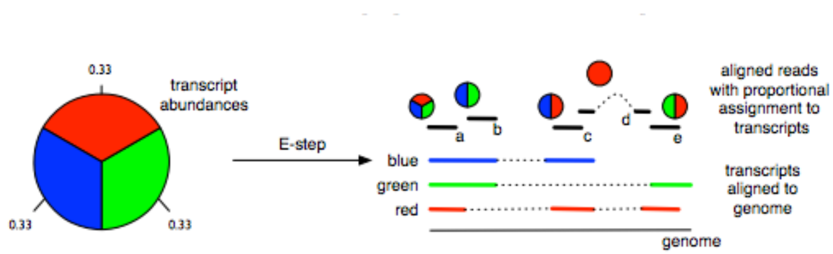
There are five reads (a,b,c,d,e) mapping to the gene.

- Read a maps to all three isoforms
- Read d only to red
- Reads b,c,e map to each of the three pairs of isoforms.

What is the most likely expression level of each isoform?

Models for transcript quantification from RNA-seq
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



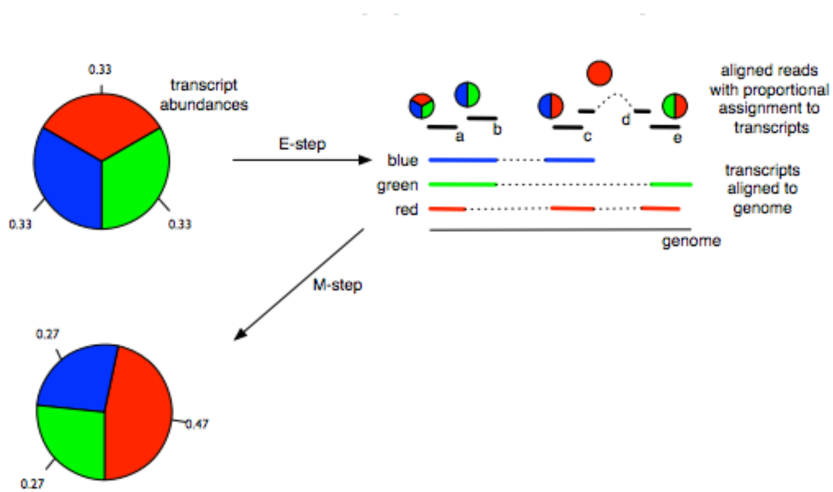
The gene has three isoforms (red, green, blue) of the same length. Initially every isoform is assigned the same abundance (red=1/3, green=1/3, blue=1/3)

There are five reads (a,b,c,d,e) mapping to the gene. Read a maps to all three isoforms, read d only to red, and the other three (reads b,c,e) to each of the three pairs of isoforms.

During the expectation (E) step reads are proportionately assigned to transcripts according to the (current) isoform abundances (RGB): a=(.33,.33,.33), b=(0,.5,.5), c=(.5,.5), d=(1,0,0), e=(.5,.5,0)

Models for transcript quantification from RNA-seq
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



The gene has three isoforms (red, green, blue) of the same length. Initially every isoform is assigned the same abundance (red=1/3, green=1/3, blue=1/3)

There are five reads (a,b,c,d,e) mapping to the gene. Read a maps to all three isoforms, read d only to red, and the other three (reads b,c,e) to each of the three pairs of isoforms.

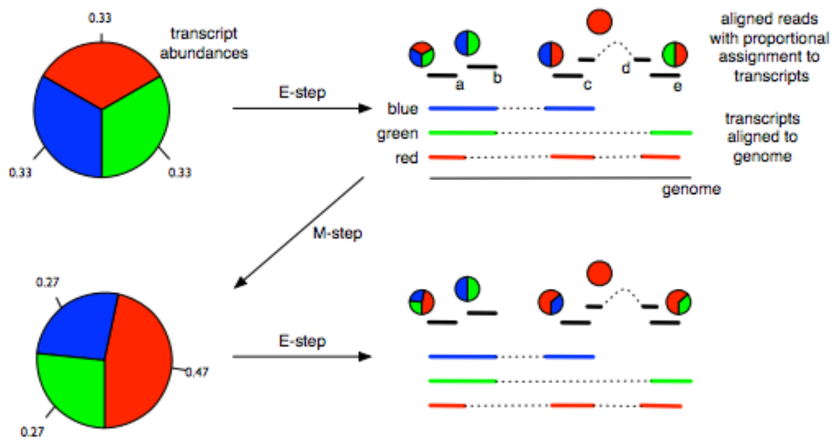
During the expectation (E) step reads are proportionately assigned to transcripts according to the (current) isoform abundances (RGB): a=(.33,.33,.33), b=(0,.5,.5), c=(.5,.5), d=(1,0,0), e=(.5,.5,0)

Next, during the maximization (M) step isoform abundances are recalculated from the proportionately assigned read counts:

red: 0.47 = (0.33 + 0.5 + 1 + 0.5)/(2.33 + 1.33 + 1.33)
blue: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)
green: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)

Models for transcript quantification from RNA-seq
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



The gene has three isoforms (red, green, blue) of the same length. Initially every isoform is assigned the same abundance (red=1/3, green=1/3, blue=1/3)

There are five reads (a,b,c,d,e) mapping to the gene. Read a maps to all three isoforms, read d only to red, and the other three (reads b,c,e) to each of the three pairs of isoforms.

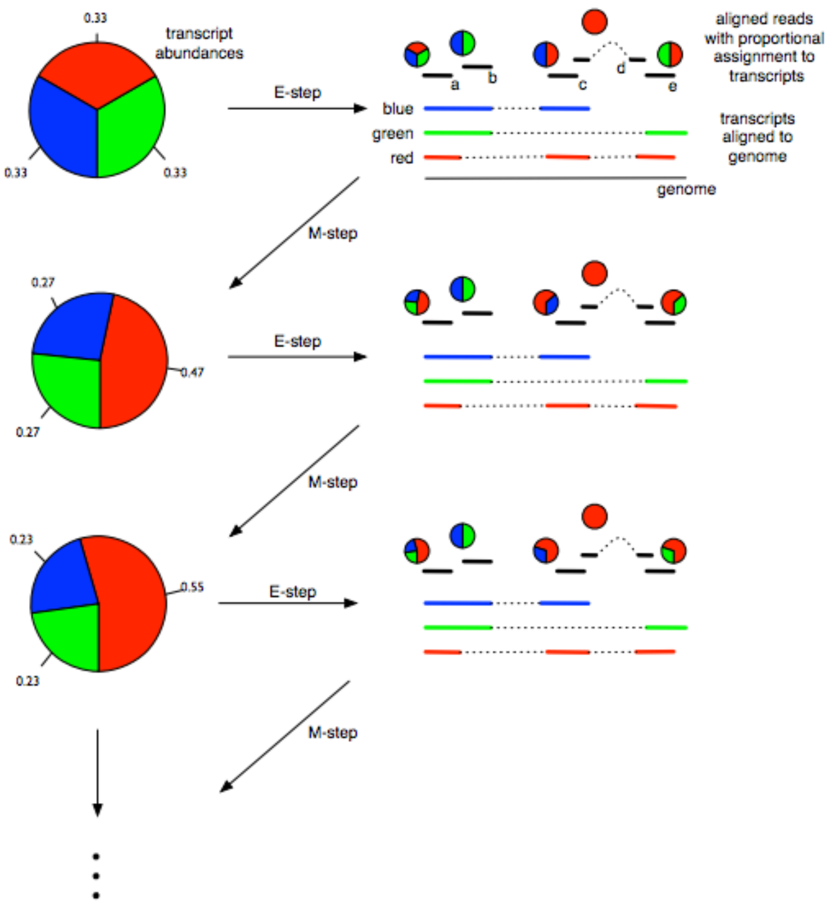
During the expectation (E) step reads are proportionately assigned to transcripts according to the (current) isoform abundances (RGB): a=(.33,.33,.33), b=(0,.5,.5), c=(.5,.5), d=(1,0,0), e=(.5,.5,0)

Next, during the maximization (M) step isoform abundances are recalculated from the proportionately assigned read counts:
red: 0.47 = (0.33 + 0.5 + 1 + 0.5)/(2.33 + 1.33 + 1.33)
blue: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)
green: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)

Repeat until convergence!

Models for transcript quantification from RNA-seq
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



The gene has three isoforms (red, green, blue) of the same length. Initially every isoform is assigned the same abundance (red=1/3, green=1/3, blue=1/3)

There are five reads (a,b,c,d,e) mapping to the gene. Read a maps to all three isoforms, read d only to red, and the other three (reads b,c,e) to each of the three pairs of isoforms.

During the expectation (E) step reads are proportionately assigned to transcripts according to the (current) isoform abundances (RGB): a=(.33,.33,.33), b=(0,.5,.5), c=(.5,.5), d=(1,0,0), e=(.5,.5,0)

Next, during the maximization (M) step isoform abundances are recalculated from the proportionately assigned read counts:

red: 0.47 = (0.33 + 0.5 + 1 + 0.5)/(2.33 + 1.33 + 1.33)
blue: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)
green: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)

Repeat until convergence!

Models for transcript quantification from RNA-seq
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]